

Zahid Razzaq

Computer Vision & Multimodal AI Researcher

Video Understanding / Action Recognition / Multimodal Learning / Human-Robot Interaction

Bolzano, Italy | Zahid.Razzaq@student.unibz.it | +39 344 474 0037

[Research Portfolio](#) | [GitHub](#) | [LinkedIn](#)

Research Profile

Final-stage PhD researcher in Robotics and Intelligent Machines specializing in **computer vision, multimodal video understanding, fine-grained action recognition, and human-robot interaction**. Develop Python/PyTorch research pipelines using RGB, Depth, gaze, and egocentric visual data, with experience adapting Video Swin Transformer/Swin3D and X3D models, designing multimodal fusion methods, conducting controlled ablations, and evaluating accuracy-complexity trade-offs. Additional experience includes Meta Project Aria data acquisition, temporal action annotation, human-participant experimentation, and EEG-based brain-robot shared control. Available for full-time research and research-engineering opportunities from **November 2026**.

Research Expertise

- **Video understanding:** fine-grained procedural action recognition, egocentric video, human activity understanding, assembly/disassembly activity analysis.
- **Multimodal learning:** RGB-Depth fusion, RGB-gaze integration, gaze-guided foveation, feature-level modulation, and early/intermediate/late/hybrid fusion.
- **Deep learning:** Python, PyTorch, Torchvision, Video Swin Transformer/Swin3D, X3D, CNNs, transformer-based video models, transfer learning, and fine-tuning.
- **Human-centred sensing:** Meta Project Aria, RGB, Depth, gaze, EEG, human-participant experiments, temporal annotation, timestamp alignment, and data-quality validation.

Research Experience

PhD Researcher – Robotics and Intelligent Machines

Feb 2023 – Oct 2026

University of Genoa, Italy; research hosted at Free University of Bozen-Bolzano

- Develop Python/PyTorch methods for fine-grained assembly and disassembly action recognition using RGB, Depth, and gaze information from egocentric and fixed-view video.
- Adapted Video Swin Transformer/Swin3D and X3D pipelines to compare **early, intermediate, late, and hybrid multimodal fusion** under consistent training and evaluation protocols.
- Designed controlled gaze-aware experiments using gaze heatmaps, explicit gaze-marker encoding, gaze-guided foveation, and feature-level modulation.
- Demonstrated an approximately **1.5 percentage-point improvement** over the controlled RGB baseline using input-level RGB-gaze integration, while also documenting failed and overfitting alternatives.
- Developed multimodal data-processing workflows including frame extraction, clip generation, spatial transformations, normalization, dataloader adaptation, model training, evaluation, and ablation analysis.
- Designed and conducted a multimodal assembly/disassembly recording study with approximately **24 participants** using Meta Project Aria and two fixed-view cameras; handled participant procedures, experimental monitoring, and data-quality checks.
- Established a reusable workflow for **fine-grained temporal action annotation** of Project Aria video, including action-boundary validation and timestamp mapping to the common device-time domain.
- Maintain reproducible research workflows using experiment configurations, checkpoints, evaluation scripts, structured logs, comparative tables, and computational-cost analysis.

Graduate Researcher – Brain-Robot Interfaces

Sep 2016 – Jan 2023

Beijing Institute of Technology, China

- Collected and processed EEG data from human participants and trained **SVM classifiers** to translate user intentions into robot-control commands.
- Developed EEG-driven shared-control methods for a brain-controlled mobile robot in MATLAB/Simulink by combining machine-learning classification with autonomous safety control.
- Designed fuzzy and self-learning neuro-fuzzy controllers for intention tracking, disturbance handling, and obstacle avoidance in human-in-the-loop simulation experiments.
- Published the resulting research in *Sensors* and the Chinese Control Conference.

Lecturer

Feb 2016 – Sep 2016

Chenab College of Engineering and Technology, Pakistan

- Taught undergraduate engineering courses and supervised laboratory and project-based activities, supporting students in implementation, experimentation, and engineering problem solving.

Selected Research Projects

Multimodal Learning for Procedural Video Understanding

- Built a unified benchmark comparing RGB–Depth fusion strategies using consistent training and evaluation protocols and analysed recognition accuracy together with latency, GFLOPs, and parameter count.

Gaze-Aware RGB–Depth Action Recognition

- Evaluated RGB+gaze integration, explicit gaze-marker encoding, gaze-guided foveation, and feature-level modulation through controlled ablations.

Multimodal Assembly/Disassembly Dataset Workflow

- Designed human-participant recording, Project Aria extraction, dataset organization, temporal action annotation, timestamp mapping, and validation workflows for downstream multimodal modelling.

Brain-Controlled Mobile-Robot Shared Control

- Integrated EEG-based SVM intention classification with fuzzy/neuro-fuzzy shared control and autonomous safety logic in MATLAB/Simulink simulation.

Selected Publications

- **Z. Razzaq et al.**, “A Comparative Evaluation of RGB–Depth Fusion for Egocentric Assembly Action Recognition,” accepted for presentation at *IC2IE 2026*.
- **Z. Razzaq**, N. Brahimi, H. Z. Ur Rehman, and Z. H. Khan, “Intelligent Control System for Brain-Controlled Mobile Robot Using Self-Learning Neuro-Fuzzy Approach,” *Sensors*, 24(18), 5875, 2024. doi:10.3390/s24185875.
- **Z. Razzaq** and L. Bi, “Fuzzy-Based Shared Control for Brain-Controlled Mobile Robot,” *39th Chinese Control Conference*, pp. 3759–3764, 2020.
- N. Brahimi, H. Zhang, and **Z. Razzaq**, “Explainable Spatio-Temporal Inference Network for Car-Sharing Demand Prediction,” *ISPRS International Journal of Geo-Information*, 14(4), 163, 2025.
- N. Brahimi, H. Zhang, and **Z. Razzaq**, “Quantum-Inspired Spatio-Temporal Inference Network for Sustainable Car-Sharing Demand Prediction,” *Sustainability*, 17(11), 4987, 2025.

Education

PhD Programme in Robotics and Intelligent Machines

University of Genoa, Italy

Research hosted at the Free University of Bozen–Bolzano. Formal defence and degree award expected by March 2027. *Research focus: multimodal action recognition, RGB–Depth/gaze fusion, egocentric video understanding, human activity understanding, and HRC.*

MSc in Mechatronics and Control Engineering

University of Engineering and Technology Lahore, Pakistan

Relevant background: artificial intelligence, robotics, signal processing, industrial automation, instrumentation, and control systems.

BSc in Mechatronics and Control Engineering

University of Engineering and Technology Lahore, Pakistan

Final project: camera-based intruder detection and two-direction motorized tracking using OpenCV/C++.

Technical Skills

Programming/ML:	Python, PyTorch, Torchvision, NumPy, pandas, Scikit-learn, TensorFlow/Keras, MATLAB/Simulink, OpenCV
Vision/Video:	Action recognition, video understanding, egocentric vision, RGB–Depth learning, gaze-guided vision, frame/clip processing
Models:	Video Swin Transformer/Swin3D, X3D, CNNs, transformer-based video models, transfer learning, fine-tuning
Multimodal/Data:	RGB, Depth, gaze, EEG, Meta Project Aria, multimodal acquisition, temporal annotation, synchronization support, data-quality validation
Evaluation:	Controlled benchmarking, ablation studies, Top-1/Top-5 accuracy, precision, recall, weighted/macro F1, latency, GFLOPs, parameter-count analysis, failure/overfitting analysis

Languages: Urdu (native) | English (professional working proficiency) | Italian (beginner)